

MAXIME JOLLIOT

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Applied ML & Autonomous Systems · Embedded AI & Perception

EDUCATION

ISAE-ENSMA | Institut Supérieur de l'Aéronautique et de l'Espace — Poitiers, France 2025 – 2028 (expected)
M.Sc. in Engineering — Specialization: Computer Science & Data (Applied ML & Autonomous Systems); Minor: Energy & Aeronautics
Relevant coursework: Machine Learning & Data Science, Numerical Methods, Signals & Linear Systems, Fluid & Thermal Mechanics, Structural Mechanics, Materials Science.

Université Bourgogne Europe | Dijon, France 2021 – 2025
B.Sc. in Physics — Highest Honors; ranked 1st of class (valedictorian, 1/23).

Lycée privé Les Arcades | Dijon, France 2018 – 2021
Vocational Baccalaureate in Sales — with Honors.

TECHNICAL SKILLS

Programming: Python (NumPy, Pandas, scikit-learn, Matplotlib), C / C++ (C++17, STL), MATLAB
ML & Computer Vision: YOLOv8 (Ultralytics), ByteTrack, OpenCV (ArUco, homography); PyTorch; TensorRT
Embedded & Robotics: NVIDIA Jetson, CUDA, GStreamer, Raspberry Pi, Pixhawk, ArduPilot / MAVLink, UART & UDP protocols
Tools: Linux, Git, socat, LTspice
Certifications: MITx MicroMasters — Mathematical Methods for Quantitative Finance (15.455x)
Languages: French (native), English (TOEIC 850), Chinese (beginner)

ENGINEERING PROJECTS

Dassault UAV Challenge — Autonomous Perception & Mission Stack Jury's Favorite Award · 2026
• Built a real-time target detection and tracking pipeline (YOLOv8 + ByteTrack) for a team-built UAV.
• Implemented pixel-to-GPS coordinate mapping (nadir camera geometry) feeding an autonomous mission layer on ArduPilot/MAVLink.
• Led the perception & navigation software; validated the full stack in SITL simulation before flight.

ENSMASTEEL — Autonomous Mobile Robot | Eurobot (French Robotic Cup) · C++17, Python 2025 – 2026
• Designed a three-layer perception-to-decision architecture: Jetson (vision + strategy) → Raspberry Pi stateless dispatcher → UART slave boards.
• Implemented real-time ArUco localization with pixel-to-millimeter homography and a finite-state strategy runner using dual-source (odometry + vision) arrival confirmation.
• Built a hardware-in-the-loop simulator (socat virtual serial + board fakes) enabling full strategy regression testing with no hardware.

Embedded-AI Drone — Personal Build | NVIDIA Jetson Orin Nano, Pixhawk 2025 – 2026 · V1 flown
• Built an end-to-end embedded-AI drone (DJI F450, Jetson Orin Nano, Pixhawk over MAVLink, CSI camera): real-time onboard YOLOv8 + ByteTrack perception on a CUDA/GStreamer pipeline, with TensorRT edge-inference optimization.
• Engineered the full data pipeline capture → annotation → training → onboard deployment; reworked low-level ArduPilot mission code (waypoint generation) and achieved the drone's first autonomous flight (V1).
• Extending the decision/control layer from a geometric finite-state machine toward imitation / policy learning (sim-to-real).

COMPUTATIONAL & QUANTITATIVE PROJECTS

Quantitative Finance | MITx MicroMasters (15.455x) · R, Python 2024 – 2025
• Tested the Random-Walk / Weak-Form Efficient Market Hypothesis on ~30 years of MSFT prices (1988–2017) using the Lo-MacKinlay variance-ratio test; modeled return dynamics with AR/MA processes (ACF/PACF).
• Built Monte Carlo portfolio simulations (binomial profitable-days, geometric waiting-time), validating theoretical mean/variance and Gaussian convergence (Central Limit Theorem).

Computational Physics — Monte Carlo & Stochastic Simulation · Python, MATLAB 2023 – 2025
• 2D Ising model of ferromagnetism (Metropolis Monte Carlo): recovered the Curie transition ($T_c \approx 2.27$) and matched Onsager's analytical magnetization curve.
• 2D Brownian motion via random walks: estimated the diffusion coefficient from ensemble- and time-averaged mean-square displacement, converging to theory ($D \approx 0.25$).

RESEARCH EXPERIENCE

- Laboratoire Interdisciplinaire Carnot de Bourgogne (ICB) | Research InternJan 2025
- Implemented the GRAPE algorithm in MATLAB for optimal two-level qubit control, reducing the cost function by up to 92%.
 - Conducted critical code review and studied physical parameters affecting quantum fidelity and energy efficiency; built transferable numerical simulation tools.

ADDITIONAL EXPERIENCE

- Private Tutor | Independent2024 – 2025
- Mentored 5 adults in career transition across mathematics, physics, methodology and work organization; guided 2 into competitive medical (PASS) programs.
- Conforama, Dijon | Sales Associate & Warehouse Operative2022 – 2024
- Furniture & bedding sales consultation with stock management, quality control and maintenance; developed customer communication and composure under pressure.

AWARDS & LEADERSHIP

- Jury’s Favorite Award — Dassault UAV Challenge 2026.
- Valedictorian & Highest Honors — B.Sc. Physics (ranked 1st of class).
- Treasurer — ENSMASTEEL robotics association (equipment, budget & logistics).
- Class & Year Representative — across vocational and bachelor’s studies (2018–2025).